Backend Technology Stack

Backend Configuration :

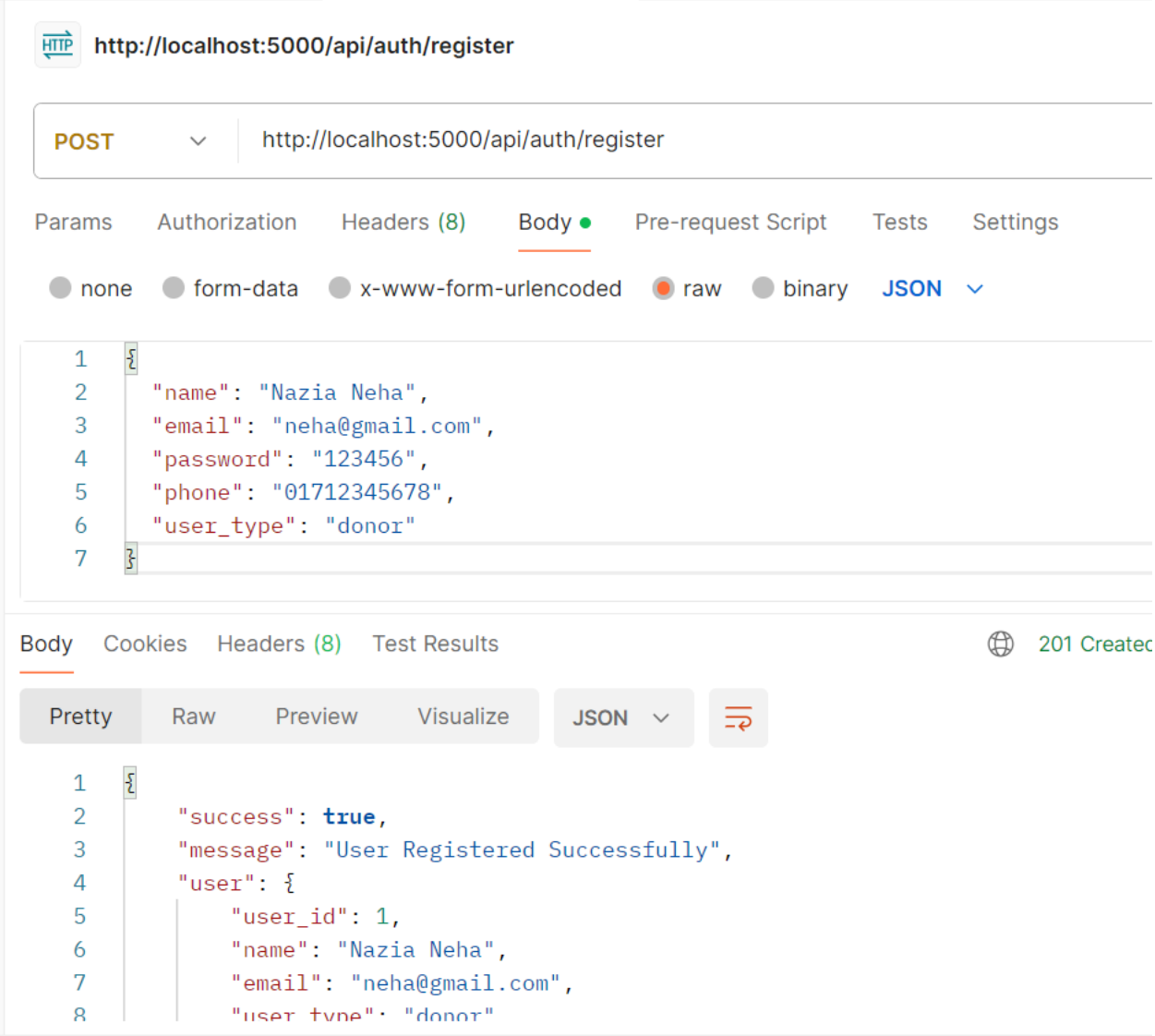
```
const { Pool } = require("pg");
require("dotenv").config();

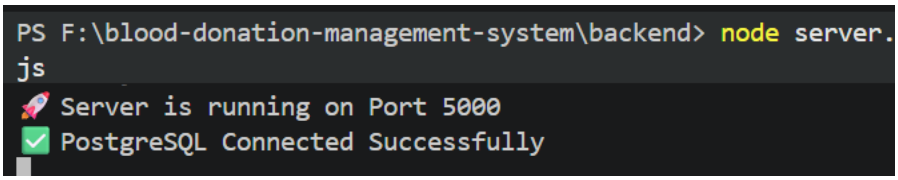
const pool = new Pool({
  host: process.env.DB_HOST,
  port: process.env.DB_PORT,
  database: process.env.DB_NAME,
  user: process.env.DB_USER,
  password: process.env.DB_PASSWORD,
});

pool.connect()
  .then(() => {
    console.log("☑ PostgreSQL Connected Successfully");
  })
  .catch((err) => {
    console.log("❌ Database Connection Failed");
    console.error(err.message);
  });

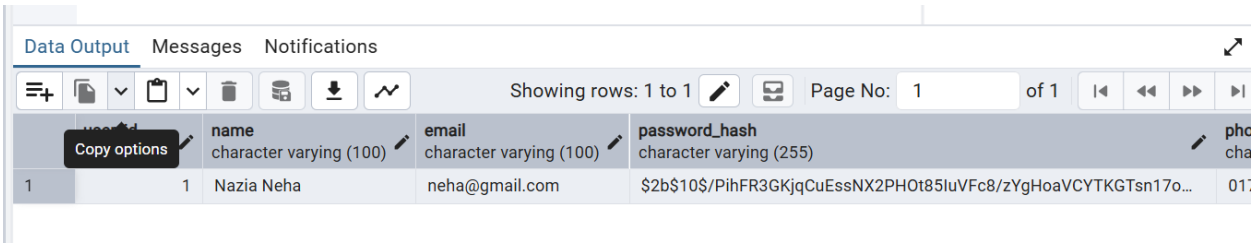
module.exports = pool;
```

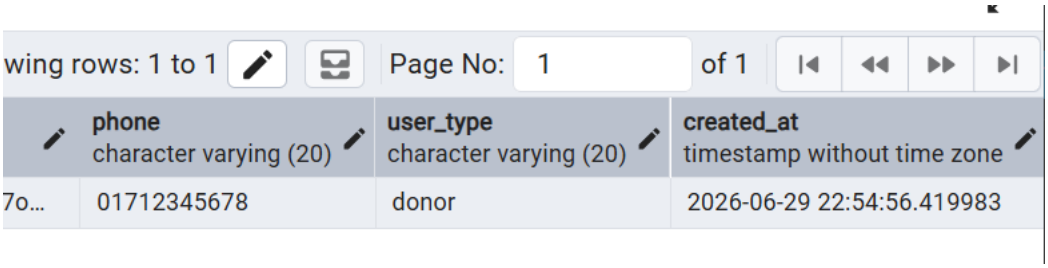
Database:





Implemented APIs :





Authentication Process:

```
const pool = require("../config/db");
const bcrypt = require("bcrypt");

const register = async (req, res) => {
    try {
        const { name, email, password, phone, user_type } = req.body;

        // Check if email already exists
        const existingUser = await pool.query(
            "SELECT * FROM users WHERE email = $1",
            [email]
        );

        if (existingUser.rows.length > 0) {
            return res.status(400).json({
                success: false,
                message: "Email already exists"
            });
        }

        // Password Hash
        const hashedPassword = await bcrypt.hash(password, 10);

        // Insert User
        const result = await pool.query(
```

```
            `INSERT INTO users(name,email,password_hash,phone,user_type)
             VALUES($1,$2,$3,$4,$5)
             RETURNING user_id,name,email,user_type`,
            [name, email, hashedPassword, phone, user_type]
        );

        res.status(201).json({
            success: true,
            message: "User Registered Successfully",
            user: result.rows[0]
        });

    } catch (error) {
        console.error(error);

        res.status(500).json({
            success: false,
            message: "Server Error"
        });
    }
};

module.exports = { register };
```

Router :

```
const express = require("express");
const router = express.Router();

const { register } = require("../controllers/authController");

router.post("/register", register);

module.exports = router;
```

Server.js :

```
const express = require("express");
const cors = require("cors");
require("dotenv").config();
```

```
require("./config/db");

const authRoutes = require("./routes/authRoutes");

const app = express();

app.use(cors());
app.use(express.json());

// Default Route
app.get("/", (req, res) => {
    res.send("AI Blood Donation Management System Backend is Running...");
});

// Authentication Routes
app.use("/api/auth", authRoutes);

// Start Server
const PORT = process.env.PORT || 5000;

app.listen(PORT, () => {
    console.log(`🚀 Server is running on Port ${PORT}`);
});
```

Backend Folder Structure :

